THE SHEAF OF MAPS OUT OF A REPRESENTABLE SHEAF IS A PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The adjunction between the product and the sheaf of maps. Same chapter, Lemma lemma-internal-hom-sheaf. Rewrites morphisms into the sheaf of maps as morphisms from a product.
- The restriction and lower shriek along a localization. Same chapter, Lemma lemma-compute-j-shriek-restrict. Identifies the lower shriek of the restriction with the product with the representable sheaf.
- The Yoneda lemma. Categories, standard form of the Yoneda lemma. Turns the chain of bijections into an isomorphism of sheaves.

Source reference. Original lemma 0BWQ: sources/stacks-sites/pinned-original-source.tex, source lines 6029–6029. Navigation pointer invoked at proof line 6029; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EE: sources/stacks-sites/pinned-original-source.tex, source lines 6030–6030. Navigation pointer invoked at proof line 6030; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $\mathcal{C}$ be a site and $U \in \mathrm{Ob}(\mathcal{C})$. Then $\mathcal{H}om(h_U^\#, \mathcal{F}) = j_*(\mathcal{F}|_{\mathcal{C}/U})$ for $\mathcal{F}$ in $Sh(\mathcal{C})$.*

Proof. This can be shown by directly constructing an isomorphism of sheaves. Instead we argue as follows. Let $\mathcal{G}$ be a sheaf on $\mathcal{C}$. Then

$$\begin{aligned} \mathrm{Mor}(\mathcal{G}, j_*(\mathcal{F}|_{\mathcal{C}/U})) &= \mathrm{Mor}(\mathcal{G}|_{\mathcal{C}/U}, \mathcal{F}|_{\mathcal{C}/U}) \\ &= \mathrm{Mor}(j_!(\mathcal{G}|_{\mathcal{C}/U}), \mathcal{F}) \\ &= \mathrm{Mor}(\mathcal{G} \times h_U^\#, \mathcal{F}) \\ &= \mathrm{Mor}(\mathcal{G}, \mathcal{H}om(h_U^\#, \mathcal{F})) \end{aligned}$$

and we conclude by the Yoneda lemma. Here we used Lemmas 0BWQ and 03EE. □